

CHUHAN ZHANG

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RESEARCH INTERESTS

Dynamic robots across locomotion and manipulation, at the intersection of *model-based control* and *reinforcement learning*. As a full-stack roboticist, I build complete systems, from mechanism and embedded real-time control to sim-to-real learning, making robots more robust and agile.

EDUCATION

- **Johns Hopkins University** Aug. 2026 – Dec. 2027 (Expected)
M.S.E. in Robotics (Medical Robotics track)
- **Technion – Israel Institute of Technology (GTIIT)** Sep. 2022 – Jul. 2026
B.S. in Mechanical Engineering and Robotics (GPA: 88.0/100, $\approx 3.7/4.0$)
- **The Chinese University of Hong Kong (CUHK)** Jul. 2025 – Sep. 2025
Visiting Research Student

PUBLICATIONS

1. **Chuhan Zhang**, Hongbo Zhang, Yanlin Chen, Yunxi Tang, Yun-Hui Liu, Mingyi Liu, Xiangyu Chu. Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation. IEEE International Conference on Automation Science and Engineering (CASE), 2026. (Accepted; Best Student Paper Award Finalist) [[arXiv](#)] [[Video](#)]
2. Tongchen Lin[†], Yanqiu Zheng[†], **Chuhan Zhang**, Ruigang Chen, Yizhar Or, Mingyi Liu. Dynamics, Stability, and Energy Efficiency of an Energy-Recycling Rimless Wheel with Spring-Clutch Legs. Robotica, 2026. (Under review) [[arXiv](#)] [[Video](#)]

RESEARCH EXPERIENCE

- Propeller-Assisted 3D Hopping Robot** Jul 2025 – Present
Research Assistant, CUHK & GTIIT Advisor: Prof. Xiangyu Chu (CUHK)
Model-based control (first author, CASE 2026): Jul 2025 – Mar 2026
- Built Pro-OMEGA, a propeller-assisted 3D monopedal hopping robot, pairing an active 3-RSR parallel leg with a trunk-mounted tri-rotor.
 - Proposed a single-rigid-body **Hierarchical Force Allocation (HFA)** controller: the leg produces the stance wrench via a friction- and torque-constrained **QP**, whereas the tri-rotor supplies the residual attitude moment in stance and full roll/pitch authority in flight.
 - Built the **real-time control stack** (500 Hz LCM loop, 3-RSR inverse kinematics, Raibert-style foot placement) on a Raspberry Pi and flight controller.
 - Validated on hardware with one fixed controller: continuous 3D hopping at ~ 1.6 m/s with **abrupt start-stop** and velocity reversals, a grass-to-wood terrain transition, and recovery from impulsive pushes within 1–3 hops.
- Reinforcement learning:** Jul 2025 – Present
- Trained **PPO** hopping policies for the 3-RSR parallel leg in **Isaac Gym** and **MuJoCo MJX** across thousands of GPU-parallel environments.
 - Compared an equivalent serial-leg model and a native closed-chain MJX model for **sim-to-real** of the closed kinematic chain.
 - Designed an **encoder-decoder policy** for flight-phase underactuation, added **domain randomization** and **actuator-delay modeling**, and benchmarked against the model-based QP controller on real-robot logs.
- Hop-wheel platform and pedipulation:** Mar 2026 – Present

- Extended Pro-OMEGA into a hop-wheel platform that switches between hopping and wheeled driving both at rest and while moving, with commanded hop-height tracking from 1 to 10 cm and mid-run retargeting, and characterized the failure modes that bound the switching envelope (insufficient hop height, rollover).
- Used the leg itself as a manipulator (pedipulation) for wall-button pressing and cart pushing, closed on an onboard RGB-D camera.
- Demonstrated a continuous outdoor mission on one platform: driving on flat ground in wheeled mode, transforming into the hopper to cross grass, then returning to flat ground to push a cart.

Energy-Efficient Legged Robot via Energy Recycling

Nov 2023 – Jul 2025

Research Assistant, GTIIT

Advisor: Prof. Mingyi Liu

- Modeled **hybrid dynamics** of a passive walker using **Lagrangian methods** and analyzed stability with **Poincaré maps** via MATLAB.
- Designed and prototyped a **spring-clutch mechanism** to recycle collision energy, achieving lower energy loss.
- Performed **motion-capture experiments** to validate the model, achieving a **60% reduction in transport cost** to 0.02.

COMPETITION EXPERIENCE

RoboMaster University League (RMUL)

Oct 2024 – Mar 2025

Mechanical Adviser for Kongfu Team, Technion, Shantou, China

- Co-founded and developed Technion, Shantou, China's first robotics club, and designed and developed the RoboMaster Standard Robot.
- Constructed a robot with omnidirectional mobility, slope-traversing suspension, and an integrated gimbal system.
- Developed a visual automatic targeting system for the gimbal firing mechanism.
- Led the team to win Second Prize in the China RoboMaster University League.

VEX-EDR Competition & Innovation Competition

Sep 2015 – Dec 2020

Team Leader, Fuzhou No. 8 Middle School, China

- Co-founded the middle school Team 9698B, leading the design, build, and programming of a holonomic ball-intake and flywheel-shooter robot for the VEX "Nothing But Net" season.
- Founded the high school Team 96666A and mentored two new members through CAD, fabrication, and autonomous control to develop a cube-stacking tray robot for the VEX "Tower Takeover" season.

HONORS & AWARDS

- Second Prize, China RoboMaster University League, 2025.
- GTIIT Dean's List Academic Scholarship, 2022–2026 (top 15%).
- Gold Award, VEX Asia-Pacific Robotics Championship (High School Group), 2019.
- China Youth Science and Technology Innovation Award, 2018.
- Champion, VEX Robotics World Championship (Middle School Group), 2016.

TECHNICAL SKILLS

Programming: C++, Python, MATLAB.

Tools: Isaac Gym, MuJoCo MJX, ROS2, LCM, SolidWorks, AutoCAD, Blender, Adobe Illustrator.

Test Scores: TOEFL iBT: 110/120.